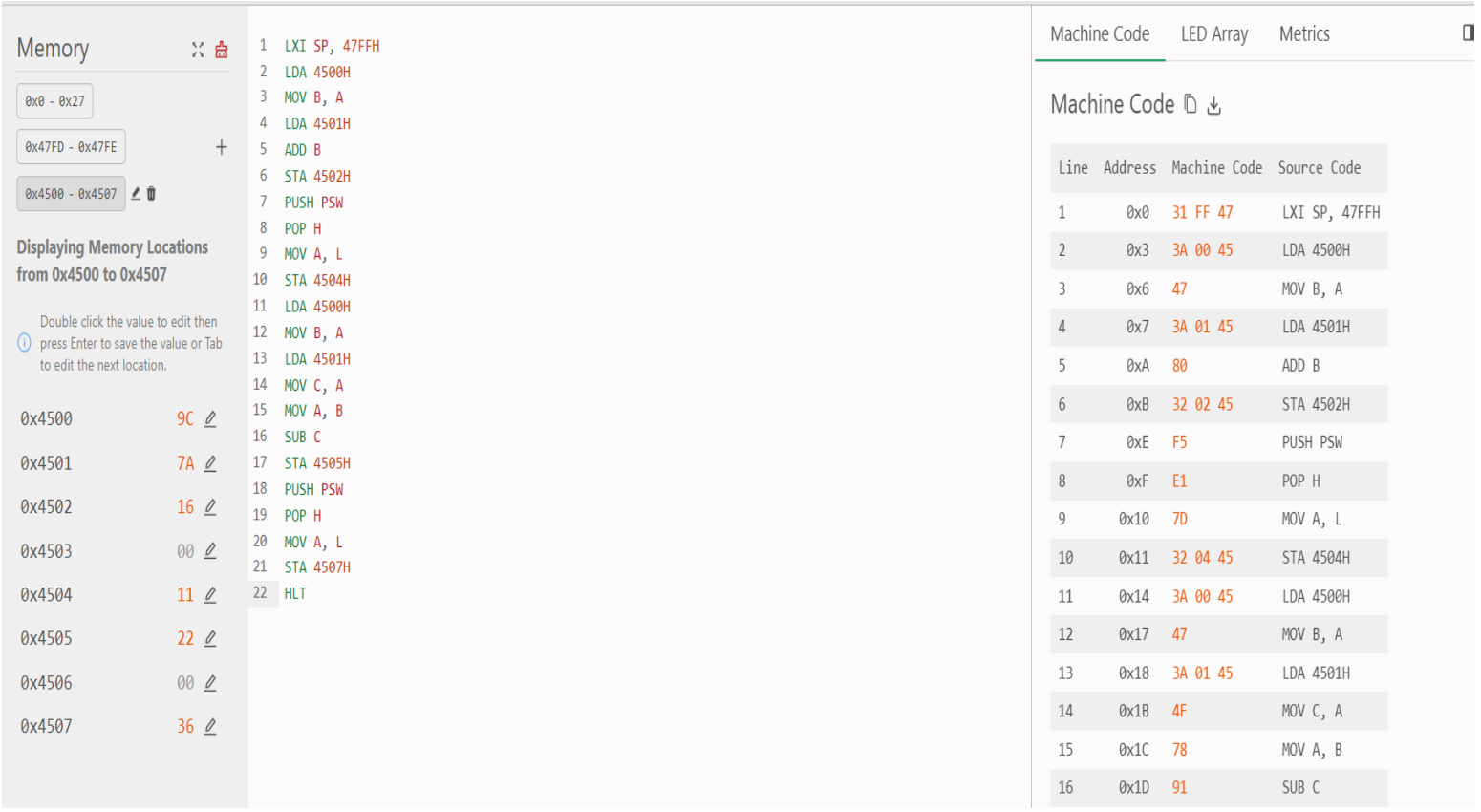
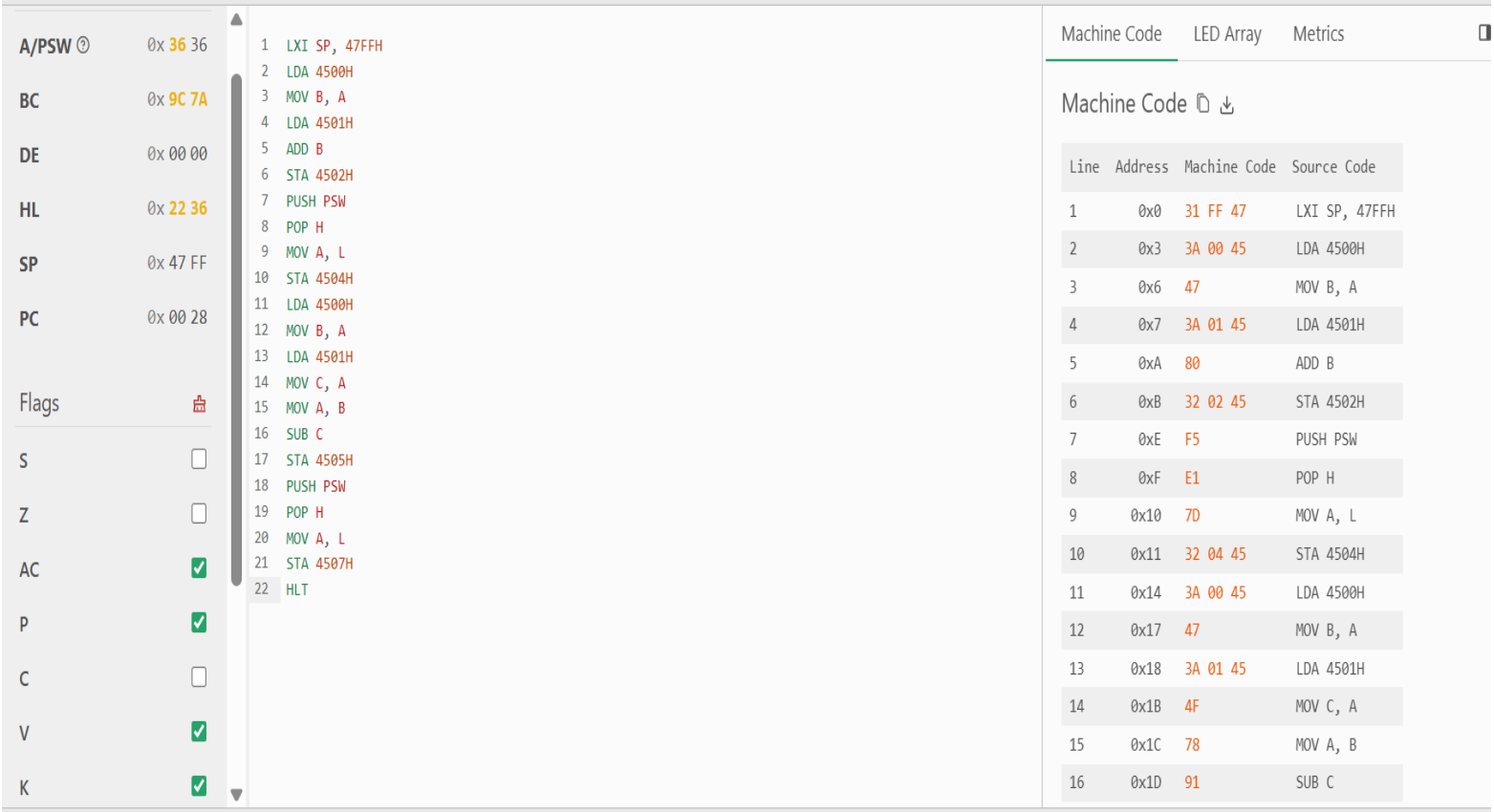
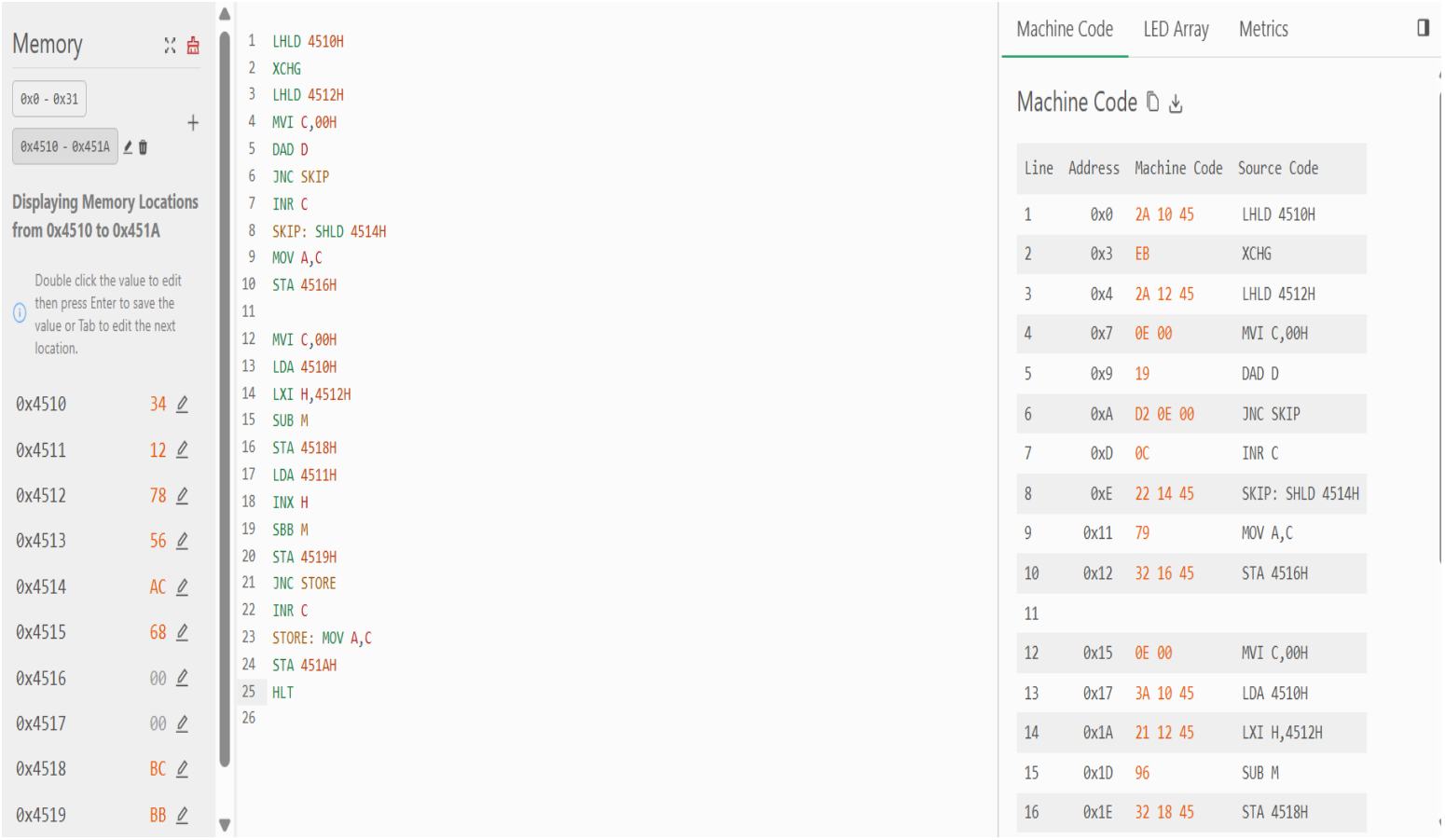
a. EXPERIMENT 1

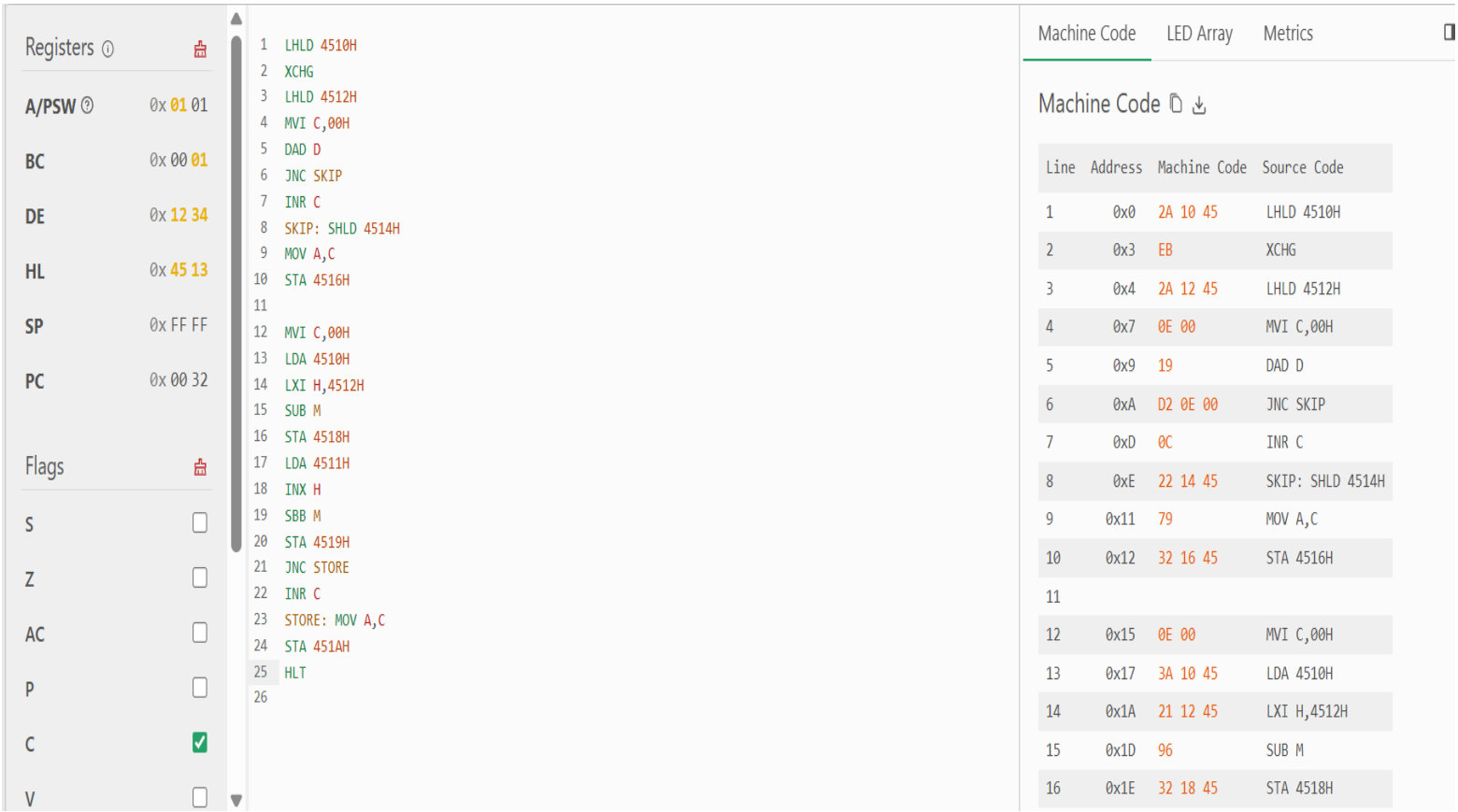
PROGRAM A :





PROGRAM B:


Memory
0x0 - 0x31
0x4510 - 0x451A
Displaying Memory Locations from 0x4510 to 0x451A
Double click the value to edit then press Enter to save the value or Tab to edit the next location.
0x4510 34
0x4511 12
0x4512 78
0x4513 56
0x4514 AC
0x4515 68
0x4516 00
0x4517 00
0x4518 BC
0x4519 BB
1 LHLD 4510H
2 XCHG
3 LHLD 4512H
4 MVI C,00H
5 DAD D
6 JNC SKIP
7 INR C
8 SKIP: SHLD 4514H
9 MOV A,C
10 STA 4516H
11
12 MVI C,00H
13 LDA 4510H
14 LXI H,4512H
15 SUB M
16 STA 4518H
17 LDA 4511H
18 INX H
19 SBB M
20 STA 4519H
21 JNC STORE
22 INR C
23 STORE: MOV A,C
24 STA 451AH
25 HLT
26
Machine Code
LED Array
Metrics
Machine Code
Line Address Machine Code Source Code
1 0x0 2A 10 45 LHLD 4510H
2 0x3 EB XCHG
3 0x4 2A 12 45 LHLD 4512H
4 0x7 0E 00 MVI C,00H
5 0x9 19 DAD D
6 0xA D2 0E 00 JNC SKIP
7 0xD 0C INR C
8 0xE 22 14 45 SKIP: SHLD 4514H
9 0x11 79 MOV A,C
10 0x12 32 16 45 STA 4516H
11
12 0x15 0E 00 MVI C,00H
13 0x17 3A 10 45 LDA 4510H
14 0x1A 21 12 45 LXI H,4512H
15 0x1D 96 SUB M
16 0x1E 32 18 45 STA 4518H


Registers
A/PSW 0x 01 01
BC 0x 00 01
DE 0x 12 34
HL 0x 45 13
SP 0x FF FF
PC 0x 00 32
Flags
S
Z
AC
P
C
V
1 LHLD 4510H
2 XCHG
3 LHLD 4512H
4 MVI C,00H
5 DAD D
6 JNC SKIP
7 INR C
8 SKIP: SHLD 4514H
9 MOV A,C
10 STA 4516H
11
12 MVI C,00H
13 LDA 4510H
14 LXI H,4512H
15 SUB M
16 STA 4518H
17 LDA 4511H
18 INX H
19 SBB M
20 STA 4519H
21 JNC STORE
22 INR C
23 STORE: MOV A,C
24 STA 451AH
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Machine Code
LED Array
Metrics
Machine Code
Line Address Machine Code Source Code
1 0x0 2A 10 45 LHLD 4510H
2 0x3 EB XCHG
3 0x4 2A 12 45 LHLD 4512H
4 0x7 0E 00 MVI C,00H
5 0x9 19 DAD D
6 0xA D2 0E 00 JNC SKIP
7 0xD 0C INR C
8 0xE 22 14 45 SKIP: SHLD 4514H
9 0x11 79 MOV A,C
10 0x12 32 16 45 STA 4516H
11
12 0x15 0E 00 MVI C,00H
13 0x17 3A 10 45 LDA 4510H
14 0x1A 21 12 45 LXI H,4512H
15 0x1D 96 SUB M
16 0x1E 32 18 45 STA 4518H

PROGRAM C :

Memory

0x0 - 0x3F

0x4520 - 0x4527

Displaying Memory Locations from 0x4520 to 0x4527

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4520	12
0x4521	05
0x4522	5A
0x4523	00
0x4524	64
0x4525	05
0x4526	14
0x4527	00

```
1  LDA 4520H
2  MOV E,A
3  MVI D,00H
4  LDA 4521H
5  MOV C,A
6  LXI H,0000H
7  ORA A
8  JZ STORE
9  LOOP: DAD D
10 DCR C
11 JNZ LOOP
12 STORE: SHLD 4522H
13
14 LDA 4525H
15 ORA A
16 JZ DIVZERO
17 MOV B,A
18 LDA 4524H
19 MVI C,00H
20 NEXT: CMP B
21 JC DONE
22 SUB B
23 INR C
24 JMP NEXT
25 DONE: STA 4527H
26 MOV A,C
27 STA 4526H
28 HLT
29
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 20 45	LDA 4520H
2	0x3	5F	MOV E,A
3	0x4	16 00	MVI D,00H
4	0x6	3A 21 45	LDA 4521H
5	0x9	4F	MOV C,A
6	0xA	21 00 00	LXI H,0000H
7	0xD	B7	ORA A
8	0xE	CA 16 00	JZ STORE
9	0x11	19	LOOP: DAD D
10	0x12	0D	DCR C
11	0x13	C2 11 00	JNZ LOOP
12	0x16	22 22 45	STORE: SHLD 4522H
13			
14	0x19	3A 25 45	LDA 4525H
15	0x1C	B7	ORA A
16	0x1D	CA 37 00	JZ DIVZERO

Registers

A/PSW	0x 14 A1
BC	0x 05 14
DE	0x 00 12
HL	0x 00 5A
SP	0x FF FF
PC	0x 00 37

Flags

S	☑
Z	☐
AC	☐
P	☐
C	☑
V	☐


```
1  LDA 4520H
2  MOV E,A
3  MVI D,00H
4  LDA 4521H
5  MOV C,A
6  LXI H,0000H
7  ORA A
8  JZ STORE
9  LOOP: DAD D
10 DCR C
11 JNZ LOOP
12 STORE: SHLD 4522H
13
14 LDA 4525H
15 ORA A
16 JZ DIVZERO
17 MOV B,A
18 LDA 4524H
19 MVI C,00H
20 NEXT: CMP B
21 JC DONE
22 SUB B
23 INR C
24 JMP NEXT
25 DONE: STA 4527H
26 MOV A,C
27 STA 4526H
28 HLT
29
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	3A 20 45	LDA 4520H
2	0x3	5F	MOV E,A
3	0x4	16 00	MVI D,00H
4	0x6	3A 21 45	LDA 4521H
5	0x9	4F	MOV C,A
6	0xA	21 00 00	LXI H,0000H
7	0xD	B7	ORA A
8	0xE	CA 16 00	JZ STORE
9	0x11	19	LOOP: DAD D
10	0x12	0D	DCR C
11	0x13	C2 11 00	JNZ LOOP
12	0x16	22 22 45	STORE: SHLD 4522H
13			
14	0x19	3A 25 45	LDA 4525H
15	0x1C	B7	ORA A
16	0x1D	CA 37 00	JZ DIVZERO

PROGRAM D :

Displaying Memory Locations from 0x4530 to 0x453F

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4530	05
0x4531	00
0x4532	03
0x4533	00
0x4534	0F
0x4535	00
0x4536	00
0x4537	00
0x4538	11
0x4539	00
0x453A	04
0x453B	00
0x453C	04
0x453D	00

```
1  LXI SP,47FFH
2  LHLD 4530H
3  XCHG
4  LHLD 4532H
5  MOV B,H
6  MOV C,L
7  LXI H,0000H
8  SHLD 4536H
9  MOV A,B
10 ORA C
11 JZ STORE
12 MLOOP: DAD D
13 JNC NOCY
14 PUSH H
15 LHLD 4536H
16 INX H
17 SHLD 4536H
18 POP H
19 NOCY: DCX B
20 MOV A,B
21 ORA C
22 JNZ MLOOP
23 STORE: SHLD 4534H
24 LHLD 453AH
25 XCHG
26 LHLD 4538H
27 LXI B,0000H
28 DLOOP: MOV A,L
29 SUB E
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	2A 30 45	LHLD 4530H
3	0x6	EB	XCHG
4	0x7	2A 32 45	LHLD 4532H
5	0xA	44	MOV B,H
6	0xB	4D	MOV C,L
7	0xC	21 00 00	LXI H,0000H
8	0xF	22 36 45	SHLD 4536H
9	0x12	78	MOV A,B
10	0x13	B1	ORA C
11	0x14	CA 2A 00	JZ STORE
12	0x17	19	MLOOP: DAD D
13	0x18	D2 24 00	JNC NOCY
14	0x1B	E5	PUSH H
15	0x1C	2A 36 45	LHLD 4536H
16	0x1F	23	INX H

Registers

A/PSW	0x FF A5
BC	0x 00 04
DE	0x 00 04
HL	0x 00 04
SP	0x 47 FF
PC	0x 00 4E

Flags

S	☑
Z	☐
AC	☐
P	☑
C	☑
V	☐


```
1  LXI SP,47FFH
2  LHLD 4530H
3  XCHG
4  LHLD 4532H
5  MOV B,H
6  MOV C,L
7  LXI H,0000H
8  SHLD 4536H
9  MOV A,B
10 ORA C
11 JZ STORE
12 MLOOP: DAD D
13 JNC NOCY
14 PUSH H
15 LHLD 4536H
16 INX H
17 SHLD 4536H
18 POP H
19 NOCY: DCX B
20 MOV A,B
21 ORA C
22 JNZ MLOOP
23 STORE: SHLD 4534H
24 LHLD 453AH
25 XCHG
26 LHLD 4538H
27 LXI B,0000H
28 DLOOP: MOV A,L
29 SUB E
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	2A 30 45	LHLD 4530H
3	0x6	EB	XCHG
4	0x7	2A 32 45	LHLD 4532H
5	0xA	44	MOV B,H
6	0xB	4D	MOV C,L
7	0xC	21 00 00	LXI H,0000H
8	0xF	22 36 45	SHLD 4536H
9	0x12	78	MOV A,B
10	0x13	B1	ORA C
11	0x14	CA 2A 00	JZ STORE
12	0x17	19	MLOOP: DAD D
13	0x18	D2 24 00	JNC NOCY
14	0x1B	E5	PUSH H
15	0x1C	2A 36 45	LHLD 4536H
16	0x1F	23	INX H

- PROGRAM A:

Given:

A = 9CH B = 7AH

(i) 8-BIT ADDITION

9C + 7A = 116

Sum = 16H

Carry = 1

(ii) 8-BIT SUBTRACTION

9C Difference = 22H

-7A Borrow = 0

22

- PROGRAM B:

(i) 16-bit Addition

1234H

+5678H

68ACH

Sum = 68ACH

Carry = 0

(ii) 16-BIT SUBTRACTION

1234H

-5678H

BB BCH

Difference = BB BCH

- PROGRAM C:

8-BIT MULTIPLICATION:

Given:

12H × 05H

(Decimal) 18 × 5 = 90

Product = 005AH

Lower byte = 5AH

Higher byte = 00H

90_{10} = 5AH

12H + 12H + 12H + 12H + 12H = 5AH

- 8-bit DIVISION:

64H ÷ 05H

100 ÷ 5 = 20

Quotient = 14H

Remainder 00H

64H - 05H = 5FH

5FH - 05H = 5AH

5AH - 05H = 55H

⋮

05H - 05H = 00H

PROGRAM D:

(1) Multiplication (X × Y)

Given:

X = 0005H = 5

Y = 0003H = 3

Calculation:

X × Y = 5 × 3 = 15

15_{10} = 000FH

Product = 00 00000FH

(2) Division:

Given:

Dividend

= 0011H = 17

Divisor = 0004H = 4

17 ÷ 4 = 4 remainder 1

17 = 4 × 4 + 1

= 16 + 1

∴ Quotient = 0004H

Remainder = 0001H